ASSIGNMENT-10

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BATCH-29

CODE:

```
class Layer2Contract:
    def __init__(self):
        self.balances = {}
        self.gas_fee = 0.0001  # very low gas (L2)

    def deposit(self, address, amount):
        self.balances[address] = self.balances.get(address, 0) +
amount
        print("Transaction Successful ✅")
        print("Deposited:", amount, "ETH")
        print("Gas Fee (L2):", self.gas_fee, "ETH")
        print("Status: Confirmed instantly ⚡\n")

    def withdraw(self, address, amount):
        if self.balances.get(address, 0) < amount:
            print("Transaction Failed ❌ - Insufficient Balance\n")
        else:
            self.balances[address] -= amount
            print("Transaction Successful ✅")
```

```
        print("Withdrawn:", amount, "ETH")
        print("Gas Fee (L2):", self.gas_fee, "ETH")
        print("Status: Confirmed instantly ⚡ \n")

    def show_balance(self, address):
        print("Wallet:", address)
        print("Balance:", self.balances.get(address, 0), "ETH\n")

# Execution starts here
contract = Layer2Contract()

user = "0x742d35Cc6634C0532925a3b844Bc454e4438f44e"

contract.deposit(user, 10)
contract.show_balance(user)

contract.withdraw(user, 3)
contract.show_balance(user)

contract.withdraw(user, 20)
```

OUTPUT:

File Edit Selection View Go Run ··· Blockchain

portfolio.html | ASSIGNMENT-8.PY | Assignment-9.py | exam.py | Ass-9.py | Ass-10.py | Ass-11.py

Ass-10.py > ...

```python
class Layer2Contract:
    def withdraw(self, address, amount):
            print("Status: Confirmed instantly ⚡ \n")

    def show_balance(self, address):
        print("Wallet:", address)
        print("Balance:", self.balances.get(address, 0), "ETH\n")


# Execution starts here
contract = Layer2Contract()

user = "0x742d35Cc6634C0532925a3b844Bc454e4438f44e"

contract.deposit(user, 10)
contract.show_balance(user)

contract.withdraw(user, 3)
contract.show_balance(user)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Gas Fee (L2): 0.0001 ETH
Status: Confirmed instantly ⚡

Wallet: 0x742d35Cc6634C0532925a3b844Bc454e4438f44e
Balance: 7 ETH

Transaction Failed ❌ - Insufficient Balance

PS C:\Users\ARSHA THALLAPALLY\OneDrive\Desktop\Blockchain>
```